

Bahadır Yüzbaşı, PhD
Professor of Statistics and Econometrics

Elected Member of the International Statistical Institute, 2023–present
Department of Econometrics
İnönü University, Malatya, Turkey
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OVERVIEW

Professor of Statistics and Econometrics whose research develops shrinkage, pretest and penalized estimation methods for high-dimensional, spatial and functional data. Current work focuses on statistical machine learning, functional data analysis, elastic shape analysis, reproducible R/C++ software, and applied data science problems in economics, biostatistics and social sciences.

Research areas: Machine learning; spatial statistics; functional data analysis; elastic shape analysis; shrinkage and penalized estimation; statistical inference; high-dimensional data; big data analytics.

Bibliometric snapshot: 94 publications; 35 WoS-indexed publications; 31 Scopus-indexed publications; 213 WoS citations; WoS h-index 10; 440 Scopus citations; Scopus h-index 14. Source: AVESİS, accessed September 25, 2026.

SKILLS

Professional programming in R and Python; R package development with C++/RcppArmadillo; reproducible statistical computing; SPSS-based statistical analysis; machine learning workflows; simulation studies; applied econometric and biomedical data analysis.

BRIEF RESEARCH STATEMENT

My research focuses on improving parameter estimation and predictive modelling in complex data settings. With my co-authors, I have developed and studied shrinkage, pretest and penalized estimation strategies for linear, partially linear, quantile and high-dimensional regression models. Recent work extends these ideas to robust estimation, spatial data analysis, functional data analysis and statistical machine learning.

These problems arise in computer science, social sciences, biometrics, finance, network security, social media analysis and industrial applications. They create both opportunities and methodological challenges, including noise accumulation, spurious correlations, heavy computational cost and algorithmic instability. My work addresses these challenges through theory, simulation, real-data analysis and reproducible implementation in R and C++.

I am especially interested in collaborative research that connects rigorous statistical methodology with substantive scientific questions.

APPOINTMENTS

Sessional Instructor Jan–May 2025
Simon Fraser University, Department of Statistics and Actuarial Science, Burnaby, Canada
- Applied Discrete Data Analysis STAT 475 / STAT 675

Visiting Professor July 2024–July 2025
Simon Fraser University, Department of Statistics and Actuarial Science, Burnaby, Canada
- Machine Learning for Functional Data with Prof. Jiguo Cao

Professor of Statistics 2023–present
İnönü University, Department of Econometrics, Malatya, Turkey

Visiting Associate Professor Feb–Aug 2019
The University of British Columbia, Department of Statistics, Vancouver, Canada
- Robust Penalized Estimation in Network Data with Prof. Ruben H. Zamar

Associate Professor of Statistics 2017–2023
İnönü University, Department of Econometrics, Malatya, Turkey

Visiting Assistant Professor 17 May–17 Aug 2016
Brock University, Department of Mathematics & Statistics, St. Catharines, Canada
- High–Dimensional Estimation Strategies with Prof. S. Ejaz Ahmed

Assistant Professor of Statistics 2015–2017
İnönü University, Department of Econometrics, Malatya, Turkey

Visiting International Scholar (PhD Research) 2013–2014
Brock University, Department of Mathematics & Statistics, St. Catharines, Canada
- Shrinkage and penalized estimation strategies with Prof. S. Ejaz Ahmed

Research Assistant 2009–2014
İnönü University, Department of Econometrics, Malatya, Turkey

EDUCATION

Associate Professorship Degree in Econometrics March/2024
Associate Professorship Degree in Statistics December/2017
PhD in Mathematics (In English) 2010–2014
İnönü University, Malatya, Turkey
"Penalty and non-penalty estimation strategies for linear and partially linear models"
Thesis advisors: Mehmet Güngör (İnönü University) & S. Ejaz Ahmed (Brock University)

MS in Mathematics (In Turkish) 2008–2010
Fırat University, Elazığ, Turkey
"Generalized Statistical Convergence"

BS in Mathematics 2004–2008
Fırat University, Elazığ, Turkey

SOFTWARE EXPERIENCE

Research software catalogue

1. `softridge` (version 0.3.0.1). Soft-thresholded ridge for high-dimensional linear models; fixed-tuning risk for the estimable projection, prediction SURE, cross-validation and least-squares refitting. R/C++17, RcppArmadillo. Co-developed with Mohammad Arashi. Website | Source.
2. `scaleCorrection` (version 0.1.0). Research companion to `softridge` for optimism-corrected scaling after soft thresholding, with matched Ridge-c0 tuning. R/C++17, RcppArmadillo. Co-developed with Mohammad Arashi. Website | Source.
3. `emaxProjection` (version 0.1.0). Confidence-region projection for integrated Emax responses, including boundary cases, feasible witnesses, endpoint gaps and termination diagnostics. R/C++17, RcppArmadillo. Website | Source.
4. `gwrs` (version 0.4.0.9015). Geographically weighted regression with joint predictor-subset, kernel and bandwidth selection, and local ridge, lasso, elastic-net, SCAD and MCP regularization. R/C++, RcppArmadillo and RcppParallel. Website | Source.
5. `AdaReFun` (version 0.1.2). Adaptive regional scalar-on-function regression with MCP and SCAD penalties, single- and multi-resolution estimation, and validation-based model selection. R/C++17, RcppArmadillo. Co-developed with Jiguo Cao. Website | Source.
6. `sglasso` (version 1.2.0). Scaled group lasso for collinear groupwise selection with Gaussian and binomial responses. R/C++, RcppArmadillo. Co-developed with Jiguo Cao. Website | Source. Related article: Yüzbaşı & Cao (2026), "Collinear Groupwise Selection via Scaled Group Lasso," *The American Statistician*.
7. `lambertReg` (version 0.1.11). Sparse Gaussian regression with the Lambert penalty, fixed-shape regularization paths, training-only cross-validation and numerical diagnostics. R/C++, Rcpp. Website | Source.
8. `rbridge` (version 1.0.4, CRAN). Restricted Bridge Estimation under linear coefficient restrictions. R/C++, RcppArmadillo. Co-developed with Mohammad Arashi and Fikri Akdeniz. Website | Source | CRAN. Related article: Yüzbaşı, Arashi & Akdeniz (2021), "Penalized Regression via the Restricted Bridge Estimator," *Soft Computing*, 25(13), 8401–8416.
9. `LHWL` (version 0.3.1). Lorentz hyperbolic weighted local regression and penalized learning for Gaussian and binomial responses, with tuning, diagnostics and coefficient maps. R/C++, RcppArmadillo. Co-developed with Zühal Küçükarslan Yüzbaşı. Website | Source.
10. `HDMaxShrink` (version 0.6.2). Test-guided shrinkage for high-dimensional linear models, with maximum-statistic calibration and compiled computational tools. R/C++, RcppArmadillo. Website | Source.
11. `BernoulliRuns` (version 0.1.0). Exact run-count distributions for independent Bernoulli trials with possibly varying success probabilities; probability masses, cumulative distributions, expectations and one-sided tails. R. Website | Source.
12. `penGWBR` (source version 1.4.1). Penalized geographically weighted beta regression for proportion responses, with lasso and adaptive-lasso local variable selection, spatial cross-validation, coefficient maps and diagnostics. R/C++, RcppArmadillo. Private source repository.
13. `ShapeAdaptiveEN` (version 0.1.0). Shape-adaptive elastic-net regression under skewness and heavy tails, with fitting, prediction and diagnostic tools. Python 3.13. Website | Source.

Programming and applied computing. Python programming for statistical computing, machine learning workflows, data cleaning and reproducible research pipelines. SPSS programming and applied statistical analysis for social science and medical research; lecturer for applied SPSS courses.

PROJECTS, SCHOLARS, AWARDS AND GRANTS

- "Modeling Heterogeneity in Big Spatial Data with Explainable and Supervised Machine Learning", TÜBİTAK 1001, Project Number: 126K746. Principal Investigator. Funding approved August 7, 2026.
- "Validity and Reliability Analysis of an AI-Developed Turkish Tax Compliance Scale", Inonu University Scientific Research Project, SBA-2026-4833, April 1, 2026–April 1, 2027. Researcher. Ongoing.

- "Analysis of the Effects of Corruption Perception on Political Behaviour in Turkey", Inonu University Scientific Research Project, SBA-2026-4536, February 17, 2026–February 17, 2027. Researcher. Ongoing.
- "Scaled Shrinkage Strategy for Complex Data Structures", Inonu University Scientific Research Project, SBA-2025-4340, November 13, 2025–November 13, 2026. Principal Investigator. Ongoing.
- "Machine Learning for Functional Data", TÜBİTAK 2219 International Postdoctoral Research Scholarship. Host: Simon Fraser University, Department of Statistics and Actuarial Science, Canada, 2024–2025, 12 months. Role: Coordinator. Budget: 31,200 USD.
- "Investigation of Domestic and Foreign Migration of Van Province by Geographically Weighted Regression Method", TÜBİTAK 3005 Innovative Solutions Research Projects Support Program in Social Sciences and Humanities, Project Number: 122G128, 01.10.2022–01.11.2023. Role: Coordinator. Budget: 327,137 Turkish Liras.
- "Shrinkage and Penalized Estimation Strategies", TÜBİTAK 2214-A International Research Fellowship Programme for PhD Students. Host: Brock University, Department of Mathematics and Statistics, Canada, 2013–2014, 12 months. Role: Coordinator. Budget: 15,600 USD.
- "Robust Regularization and Gaussian Graphical Models", Inonu University International Research Project, SUA-2019-1629, January 22–December 20, 2019. Principal Investigator. Research visit: The University of British Columbia, Canada, February–August 2019. Budget: 7,500 USD. Completed.
- "Pretest and Stein-type Estimation in Linear Quantile Regression Models", Inonu University Scientific Research Project, SBA-2017-946, December 12, 2017–November 21, 2018. Principal Investigator. Completed.
- "Liu-type Estimation in Linear Regression Models", Necmettin Erbakan University Scientific Research Project, Project Number: 171215001, 03.2017–12.2017. Role: Researcher.
- The Fields Institute, University of Toronto, Toronto, Canada, 2016, congress attendance grant.

BOOK

1. Ahmed, SE, Ahmed, F & **Yüzbaşı, B.**, (2023) Post-shrinkage strategies in statistical and machine learning for high dimensional data. Chapman and Hall/CRC, London. (Routledge and CRC Press of Taylor & Francis Group)
2. **Yüzbaşı, B.**, & Cetin Gorur, (2019) Dinamik Dogrusal Modeller ile Turkiye'ye Ait Gayri Safi Yurt ici Hasila Tahmini, Gece Akademi, Turkey.

INTERNATIONAL-REVIEWED JOURNAL PAPERS

1. **Yüzbaşı, B.**, & Cao, J. (2026). Collinear Groupwise Selection via Scaled Group Lasso. *The American Statistician*. Published online September 17, 2026. [SCIE; Q1] [TÜBİTAK-supported research]
2. **Yüzbaşı, B.**, Ahmed, S. E., & Liu, S. (2026). Shrinkage Estimation in Geographically Weighted Regression with Applications to Digital Platform Pricing. *Spatial Statistics*, 73, 100969. [SCIE; Q1]
3. Pehlivan, C., **Yüzbaşı, B.**, & Tuncel, G. (2026). The Role of Socio-Economic Development Level in Van Province on Migration. *Sinop Üniversitesi Sosyal Bilimler Dergisi*, 10(1), 23–54.
4. **Yüzbaşı, B.**, & Görür, Ç. (2026). Analysis of Van Province's Migration from Iran by Geographically Weighted Regression Method. *Uluslararası İlişkiler / International Relations*, 23(89), 27–49. [SSCI; Q3] [TÜBİTAK-supported research]
5. Görmüş, S., Cengiz, S., & **Yüzbaşı, B.** (2026). A Spatio-Temporal Analysis of COVID-19 Spread in Türkiye. *Forestist*, 76(1), 1–14. [ESCI; Scopus]
6. Hassan, M. K., Bayat, T., **Yüzbaşı, B.**, & Kayhan, S. (2026). Was Digitalization Effective in Enhancing Financial Inclusion in the EU-27 During the COVID–19 Pandemic? *Eastern European Economics*, 64(5), 778–794. [SSCI; Q3]
7. Al-Momani, M., **Yüzbaşı, B.**, Bataineh, M. S., Abdallah, R., & Moideenkutty, A. (2025). Shrinkage Approaches for Ridge-Type Estimators Under Multicollinearity. *Mathematics*, 13(22), 3733. [SCIE; Q1]

8. McGee, R. W., **Yüzbaşı, B.**, Kısa, Ö., & Benk, S. (2025). The Role of Emotions in Attitudes toward Ethics of Tax Evasion: An Empirical Study. *International Journal of Public Finance*, 10(2), 325–340. [TR Dizin]
9. Tuncel, G., Yılmaz, H., Göktolga, O., & **Yüzbaşı, B.** (2024). The Impact of Religious Officials on Social Life: A Research on Scale Development, Validity and Reliability. *Journal of Ilahiyat Researches*, 62(1), 86–100. [ESCI; TR Dizin]
10. Gorur, C., & **Yüzbaşı, B.** (2024). Study of the Factors Influencing Cultural Similarity in the Post-Migration Adaptation Process in the Province of Van Using the GWR Method. *Journal of Geography*, 48, 137–154. [TR Dizin]
11. **Yüzbaşı, B.**, & Gorur, C. (2023). Examining The Role of Job by Geographically Weighted Poisson Regression in The Post-Migration Adaptation Process: The Case of Van. Yüzüncü Yıl Üniversitesi Sosyal Bilimler Enstitüsü Dergisi, (Cumhuriyet Özel Sayısı), 11-27.
12. **Yüzbaşı, B.**, Asar, Y., & Ahmed, S. E. (2022). Liu-type shrinkage estimations in linear models. Statistics, 56(2), 396-420. [SCIE; Q3]
13. **Yüzbaşı, B.**, & Pala, M. (2022) Ridge regression parameter selection: Turkey's example of foreign direct investment. Journal of Statisticians: Statistics and Actuarial Sciences, 15(1), 1-18.
14. Arashi, M., Asar, Y., & **Yüzbaşı, B.** (2021). SLASSO: a scaled LASSO for multicollinear situations. Journal of Statistical Computation and Simulation, 91(15), 3170-3183. [SCIE; Q2]
15. **Yüzbaşı, B.**, Arashi, M, & Akdeniz, F. (2021). Penalized Regression via the Restricted Bridge Estimator. *Soft Computing*, 25(13), 8401-8416. [SCIE; Q3]
16. **Yüzbaşı, B.**, & Gorur, C. (2021). Examining Spatial Variability in the Association Between Some Factors and Number of Prisoners in Turkey: A GWR Analysis. Mugla Journal of Science and Technology, 7(1), 141-148.
17. **Yüzbaşı, B.**, & Ahmed, S. E. (2020). Ridge Type Shrinkage Estimation of Seemingly Unrelated Regressions And Analytics of Economic and Financial Data from "Fragile Five" Countries. *Journal of Risk and Financial Management*, 13(6), 131.
18. **Yüzbaşı, B.**, Ahmed, S. E., & Aydin, D. (2020). Ridge-type pretest and shrinkage estimations in partially linear models. *Statistical Papers*, 61(2), 869–898. [SCIE; Q3] [TÜBİTAK-supported research]
19. **Yüzbaşı, B.**, Arashi, M., & Ejaz Ahmed, S. (2020). Shrinkage Estimation Strategies in Generalised Ridge Regression Models: Low/High-Dimension Regime. *International Statistical Review*, 88(1), 229–251. [SCIE; Q1]
20. McGee, R. W., Benk, S., **Yüzbaşı, B.**, & Budak, T. (2020). Does Religiosity Affect Attitudes toward the Ethics of Tax Evasion? The Case of Turkey. Religions, 11(9), 476. [AHCI; Q1]
21. **Yüzbaşı, B.**, & Arashi, M. (2019). Double shrunken selection operator. *Communications in Statistics-Simulation and Computation*, 48(3), 666-674. [SCIE; Q3]
22. McGee, R. W., Benk, S., & **Yüzbaşı, B.**. (2019). Should Governments Tax the Rich and Subsidize the Poor? A Comparative Study of Muslim and Christian Respondents. *Religions, 10(2), 72.* [AHCI; Q1]
23. Yilmaz, E., **Yüzbaşı, B.**, & Aydin, D. (2021). Choice of smoothing parameter for kernel type ridge estimators in semiparametric regression models. *REVSTAT-Statistical Journal*, 19(1), 47–69. [SCIE; Q2]
24. **Yüzbaşı, B.**, Asar, Y., Sik, M. S., & Demiralp, A. (2018). Improving Estimations in Quantile Regression Model with Autoregressive Errors. Thermal Science. Volume 22, Issue Suppl. 1, Pages: 97-10, [SCIE; Q4]
25. Arashi, M., Norouzirad, M., Ahmed, S. E., & **Yüzbaşı, B.** (2018). Rank-based Liu regression. *Computational Statistics*, 33(3), 1525-1561. [SCIE; Q2]
26. **Yüzbaşı, B.**, Ahmed S.E, & Gungor, M. (2017). Improved Penalty Strategies in Linear Regression Models, *REVSTAT-Statistical Journal*, 15(2), 251–276. [SCIE; Q3] [TÜBİTAK-supported research]
27. Benk, S., **Yüzbaşı, B.**, & McGee, R. W. (2017). Confidence in Government and Attitudes toward Bribery: A Country-Cluster Analysis of Demographic and Religiosity Perspectives. *Religions*, 8(1), 8. [AHCI; Q1]
28. **Yüzbaşı, B.**, & Ejaz Ahmed, S. (2016). Shrinkage and penalized estimation in semi-parametric models with multicollinear data. *Journal of Statistical Computation and Simulation*, 86(17), 3543–3561. [SCIE; Q2] [TÜBİTAK-supported research]
29. **Yüzbaşı, B.**, Bulut, Y., & Gungor, M. (2016). On order statistics from nonidentical discrete random variables. *Open Physics*, 14(1), 192-196. [SCIE; Q2]

30. Aydın, D., **Yüzbaşı, B.**, & Ahmed, S. E. (2016). Modified ridge type estimator in partially linear regression models and numerical Comparisons, *Journal of Computational and Theoretical Nanoscience*, Vol. 13, 1–14.

31. Bulut, Y., Gungor, M., **Yüzbaşı, B.**, Ozbey, F., & Canpolat, E. (2016). On Distributions of Discrete Order Statistics. *Journal of Computational Analysis & Applications*, 20(1), 187-200. [SCIE; Q4]

32. Ejaz Ahmed, S., & **Yüzbaşı, B.** (2016). Big data analytics: integrating penalty strategies. *International Journal of Management Science and Engineering Management*, 11(2), 105-115.

33. Benk, S., Budak, T., **Yüzbaşı, B.**, & Mohdali, R. (2016). The Impact of Religiosity on Tax Compliance among Turkish Self-Employed Taxpayers. *Religions*, 7(4), 37. [AHCI; Q1]

34. Benk, S., McGee, R. W., & **Yüzbaşı, B.** (2015). How religions affect attitudes toward ethics of tax evasion? A comparative and demographic analysis. *Journal for the Study of Religions and Ideologies*, 14(41), 202. [AHCI; Q1]

35. McGee, R. W., Benk, S., & **Yüzbaşı, B.** (2015). Religion and Ethical Attitudes toward Accepting a Bribe: A Comparative Study. *Religions*, 6(4), 1168-1181. [AHCI; Q1]

36. Kayhan, S., Bayat, T., & **Yüzbaşı, B.** (2013). Government expenditures and trade deficits in Turkey: Time domain and frequency domain analyses. *Economic Modelling*, 35, 153-158. [SSCI; Q1]

37. **Yüzbaşı, B.**, & Gungor, M. (2013). On Joint Distributions of Order Statistics from Nonidentically Distributed Discrete Vectors. *Journal of Computational Analysis & Applications*, 15(1). [SCIE; Q4]

38. Gungor, M., Bulut, Y., & **Yüzbaşı, B.** (2012). On joint distributions of order statistics for a discrete case. *Journal of Inequalities and Applications*, 2012(1), 264.

BOOK REVIEWS

- **Yüzbaşı, B.** (2026). Review of *Using R for Introductory Statistics in Social Sciences*, by Mark A. Perkins (Springer, 2026, xiii + 207 pp., ISBN 978-3-032-17144-3). *International Statistical Review*. Published online September 30, 2026. doi:10.1111/insr.70072. [SCIE; Q1]

BOOK CHAPTERS AND CONFERENCE PROCEEDINGS

1. Gorur, C., Kucuk, B. & **Yüzbaşı, B.**, (2023), Examining the Effect of Education on the Post-Migration Adaptation Process with the Geographically Weighted Poisson Regression Method: Van Example. Globalization and Migration, Chapter 8, 159-175, Holistence Publications, Canakkale.

2. McGee, R. W., Benk, S., & **Yüzbaşı, B.** (2023). Jewish attitudes toward bribery. In The Ethics of Bribery: Theoretical and Empirical Studies (pp. 123-143). Cham: Springer International Publishing.

3. McGee, R. W., Benk, S., & **Yüzbaşı, B.** (2023). Buddhist attitudes toward bribery. In The Ethics of Bribery: Theoretical and Empirical Studies (pp. 145-165). Cham: Springer International Publishing.

4. McGee, R. W., Benk, S., & **Yüzbaşı, B.** (2023). Hindu attitudes toward bribery. In The Ethics of Bribery: Theoretical and Empirical Studies (pp. 101-121). Cham: Springer International Publishing.

5. McGee, R. W., Benk, S., & **Yüzbaşı, B.** (2023). Atheist attitudes toward bribery. In The Ethics of Bribery: Theoretical and Empirical Studies (pp. 79-99). Cham: Springer International Publishing.

6. McGee, R. W., Benk, S., & **Yüzbaşı, B.** (2023). Religious attitudes toward bribery: A comparative study. In The Ethics of Bribery: Theoretical and Empirical Studies (pp. 11-29). Cham: Springer International Publishing.

7. McGee, R. W., Benk, S., & **Yüzbaşı, B.** (2023). Christian attitudes toward bribery. In The Ethics of Bribery: Theoretical and Empirical Studies (pp. 31-55). Cham: Springer International Publishing.

8. Ahmed, S. E., & **Yüzbaşı, B.** Big data analytics: integrating penalty strategies, Big Data and Information Theory, Routledge, 2022.

9. **Yüzbaşı, B.**, Ahmed S.E., Arashi M., Norouzirad M. (2020) LAD, LASSO and Related Strategies in Regression Models. In: Xu J., Ahmed S., Cooke F., Duca G. (eds) Proceedings of the Thirteenth International Conference on Management Science and Engineering Management. ICMSEM2019. Advances in Intelligent Systems and Computing, vol 1001. Springer, Cham

10. **Yüzbaşı, B.**, Ahmed, S. E., & Asar, Y. (2018, December). L1 Correlation-Based Penalty in High-Dimensional Quantile Regression. In 2018 4th International Conference on Big Data and Information Analytics (BigDIA) (pp. 1-5). IEEE.

11. Ahmed, S. E., Hwanwoo, K., Yildirim, G. & **Yüzbaşı, B.** (2019). Matrices, Statistics and Big Data. IWMS 2016. Contributions to Statistics. Springer, Cham Chapter name: (High dimensional regression under correlated Design: an extensive simulation study) In: Ahmed S., Carvalho F., Puntanen S. (eds)

12. Asar, Y., Ahmed, S. E., & **Yüzbaşı, B.** (2018, August). Efficient and Improved Estimation Strategy in Zero-Inflated Poisson Regression Models. In International Conference on Management Science and Engineering Management (pp. 329-342). Springer, Cham.

13. **Yüzbaşı, B.** & Asar, Y. (2018). Econometrics: Methods & Applications, Gazi Kitabevi, Chapter name: Ridge Type Estimation in the Zero-Inflated Negative Binomial Regression

14. Ahmed, S. E. & **Yüzbaşı, B.** (2017) High Dimensional Data Analysis: Integrating Submodels. In Big and Complex Data Analysis (pp. 285-304). Springer International Publishing.

15. **Yüzbaşı, B.**, & Ahmed, S. E. (2015). Shrinkage Ridge Regression Estimators in High-Dimensional Linear Models. In Proceedings of the Ninth International Conference on Management Science and Engineering Management (pp. 793-807). Springer Berlin Heidelberg.

INVITED TALKS AND COMMITTEES AT INTERNATIONAL CONFERENCES

1. **Yüzbaşı, B.** (2026). The Second Sharjah International Conference on Mathematical Sciences (UOS-SICMS26), University of Sharjah, Sharjah, United Arab Emirates, January 14–15, 2026. (Invited Speaker)

2. **Yüzbaşı, B.** (2026). A Hybrid Regularization Framework for Correlated Feature Selection in High-Dimensional Data. 15th Conference on High Dimensional Data Analysis (HDDA-XV), Istanbul Medipol University, Istanbul, Türkiye, August 12–14, 2026. (Invited Speaker)

3. **Yüzbaşı, B.** (2025). Groupwise Feature Selection in High-Dimensional Data. International Conference on Statistics and Data Science, June 23–25, 2025, Vancouver, Canada. (Invited Speaker)

4. 44th Annual Meeting of the Statistical Society of Canada, Brock University, Canada, May 2016 (Invited Speaker)

5. 6th International Workshop on the Perspectives on High-Dimensional Data Analysis (HDDA-VI), The Fields Institute, University of Toronto, Canada, May 2016 (Invited Speaker and Organizing Committee)

6. L_q-type Penalty Estimation Under the Linear Restriction, The University of British Columbia, Department of Statistics, Vancouver, Canada, August 2019 Statistics Seminar

PRESENTATIONS AT INTERNATIONAL CONFERENCES

1. **Yüzbaşı, B.** (2023). Liu Estimation on Geographically Weighted Regression. IV. International Applied Statistics Congress, Sarajevo, Bosnia and Herzegovina.

2. **Yüzbaşı, B.** and Yüzbaşı Z. (2023). Minkowski Distance on Geographically Weighted Regression, IV. International Applied Statistics Congress, Sarajevo / Bosnia and Herzegovina

3. Pehlivan, C. & **Yüzbaşı, B.** & Tuncel, G. (2023). Sosyo-Gelişmişlik Düzeyi ile Göç Arasındaki İlişki: Van İli Üzerine Bir Araştırma. The Migration Conference 2023, 23-26 August 2023, Hamburg, Germany.

4. Pehlivan, C. & **Yüzbaşı, B.** (2023). Ekonomik Yetersizlik ile Göç Arasındaki İlişkinin Ampirik Analizi: Van İlinden Kanıtlar The Migration Conference 2023, 23-26 August 2023, Hamburg, Germany.

5. Küçük, B. & **Yüzbaşı, B.** (2023). Van İlinin Göç Sonrası Uyum Sürecinde Eğitim Etkisi çalışması. 2nd International Artificial Intelligence and Social Sciences Congress, Ufuk University, Ankara, Turkey.
6. Küçük, B. & **Yüzbaşı, B.** (2023). The Effect of Climate and Geography on Migration to Van Province. The Seventh International Conference on Computational Mathematics and Engineering Sciences (CMES-2023), 20–21 May 2023, Elazığ, Türkiye.
7. Karakaş, M. & **Yüzbaşı, B.** (2023). Migration to Van Province to Access a Better Life. The Seventh International Conference on Computational Mathematics and Engineering Sciences (CMES-2023), 20–21 May 2023, Elazığ, Türkiye.
8. Görür, Ç. & **Yüzbaşı, B.** (2023). Van İlinin Eğitim İmkânsızlıkları Nedeniyle Aldığı Göçün Coğrafi Ağırlıklı Poisson Regresyon Yöntemiyle İncelenmesi. Ufuk Üniversitesi Sosyal Bilimler Enstitüsü, "Yapay Zeka ve Sosyal Bilimler" ana temalı II. Uluslararası Sosyal Bilimler Kongresi, 3-5 Mayıs 2023, Ankara, Türkiye.
9. Görür, Ç. & **Yüzbaşı, B.** (2023). Investigation of Work Effect in the Post-Migration Adaptation Process by Geographically Weighted Poisson Regression: The Case of Van. The Seventh International Conference on Computational Mathematics and Engineering Sciences (CMES-2023), 20–21 May 2023, Elazığ, Türkiye.
10. Görür, Ç., **Yüzbaşı, B.** & Tuncel, G. (2023). Examining the Factors Affecting Cultural Similarity in the Post-Migration Adaptation Process by Geographically Weighted Regression Method: The Case of Van Province. The Migration Conference 2023, 23–26 August 2023, Hamburg, Germany.
11. Görür, Ç. & **Yüzbaşı, B.** (2023). Investigation of Migration due to Social Environmental Pressure by Geographical Regression: The Example of Van Province. The Migration Conference 2023, 23–26 August 2023, Hamburg, Germany.
12. **Yüzbaşı, B.** (2022). Big Data Analytics via Penalized Estimation. 6th International Conference on Computational Mathematics and Engineering Sciences.
13. **Yüzbaşı, B.**, ASAR YASİN, YUZBASI ZK, (2018), A new quantile penalty estimator, 11th International Statistics Days Conference.
14. **Yüzbaşı, B.**, DEMIRALP AHMED, GUNGOR MEHMET (2018). Correlation Based Penalty Estimator in Vector Autoregressive Model. 11th International Statistics Days Conference.
15. **Yüzbaşı, B.**, Ejaz Ahmed, ASAR YASİN, (2018), Liu Pretest and Stein-Type Shrinkage Estimation in Linear Regression Models, 11th International Statistics Days Conference.
16. ASAR YASİN, **Yüzbaşı, B.** (2018), A biased estimation method in zero inflated count models, 3rd International Conference on Computational Mathematics and Engineering Sciences
17. ASAR YASİN, **Yüzbaşı, B.** (2018), Quantile correlation based penalty estimator, 3rd International Conference on Computational Mathematics and Engineering Sciences.
18. **Yüzbaşı, B.**, ASAR YASİN (2018), Improved Biased Estimation Methods in Logistic Regression, 3rd International Conference on Computational Mathematics and Engineering Sciences.
19. **Yüzbaşı, B.**, ASAR YASİN, (2017). Pretest and Stein-Type Estimations in Quantile Regression Model. International Workshop on Mathematical Methods in Engineering, Russia
20. ASAR YASİN, **Yüzbaşı, B.** (2017). Linear Unified Shrinkage Estimators in Linear Models. International Workshop on Mathematical Methods in Engineering, Russia
21. DEMİRALP AHMET,ASAR YASİN, **Yüzbaşı, B.** (2017). Shrinkage and penalty estimation in quantile regression model with autoregressive errors. INTERNATIONAL CONFERENCE ON MATHEMATICAL MODELLING in Applied Sciences, Russia
22. ŞIK MUHAMMED ŞAMİL, **Yüzbaşı, B.**,ASAR YASİN (2017). Preliminary and shrinkage estimation in quantile regression model with iid errors. International Conference on Mathematical Modelling in Applied Sciences
23. ŞIK MUHAMMED ŞAMİL, **Yüzbaşı, B.**,ASAR YASİN (2017). Controlling Bias in Quantile Regression Models under Gauss-Markov Assumptions. 3rd International Researchers, Statisticians and Young Statisticians Congress
24. **Yüzbaşı, B.**,AHMED SYED EJAZ,ARASHI MOHAMMAD (2017). Improved Estimation Strategies in Generalized Ridge Regression Models. 3. International Researchers, Statisticians and Young Statisticians Congress

25. DEMİRALP AHMET,ASAR YASİN, **Yüzbaşı, B.** (2017). Improved Estimation Strategies in Quantile Regression Model when Errors are Non-iid. 3.International Researchers, Statisticians and Young Statisticians Congress 2017
26. YILDIRIM GOKHAN,AHMED SYED EJAZ,KIM HWANWOO, **Yüzbaşı, B.** (2016). On Lasso trade off diagram under correlated random design matrix. 6th International Workshop on the Perspectives on High-Dimensional Data Analysis (HDDA-VI), Fields Institute, Toronto University, Canada.
27. YILMAZ ERSİN, **Yüzbaşı, B.**,AYDIN DURSUN (2015). Ridge type kernel smoothing in semi parametric regression models A Comparative Simulation Study. 9. International Statistics Congress
28. **Yüzbaşı, B.** AHMED SYED EJAZ,GÜNGÖR MEHMET (2015). Shrinkage Ridge Type Estimators in Linear Models. 9. International Statistics Congress.

ADVISING THESIS

- Cetin Gorur, Gross domestic product estimation of Turkey with dynamic line models, Master Thesis, Inonu University, 2019.
- Emsal Cagla Avcu, An application on Lasso VAR and VARX models, Master Thesis, Inonu University, 2019.
- Ozlem Taya, Classification with support vector machines and logistic regression: Turkey's import and export example, Master Thesis,Inonu University, 2022
- Mustafa Pala, Ridge regression parameter selection: Turkey's foreign direct investment example, Master Thesis, Inonu University, 2022.
- Merve Karakas, Generalized Additive Models: An Application of Econometric Data, Master Thesis, Inonu University, 2024. Completed.
- Cetin Gorur, Investigation of Domestic and Foreign Migration of Van Province by Geographically Weighted Regression Method, PhD Thesis, Inonu University, 2024. Completed.

COURSE

- Applied SPSS analysis, İnönü University, Two months in 2012, As lecturer
- Applied SPSS analysis, İnönü University, Two months in 2024, As lecturer

MEMBERSHIP

- Elected Member of the International Statistical Institute (ISI), 2023–present.
- Member of the International Association for Statistical Computing (IASC), 2023–present.

WORKSHOPS

1. HDDA-VI: The Sixth International Workshop on the Perspectives on High-Dimensional Data Analysis, The Fields Institute, Toronto, May 2016. Role: Scientific and Organizing Committee.
2. Committee member of International Symposium on Big Data and Applied Statistics (ISBDAS), Guangzhou, China, November 2018.

TEACHING EXPERIENCE

I have taught undergraduate courses in experimental design, probability, categorical data analysis, mathematical statistics, regression analysis, linear algebra, numerical analysis, multivariate statistics, and introductory statistics. At the master's and PhD levels, I have taught statistical methods, machine learning and big data analytics.

From 2022–2026, I taught at least five courses per semester at İnönü University, including a course in English. I also taught Applied Discrete Data Analysis (STAT 475/675) at Simon Fraser University in spring 2025.

REVIEWS

1. Uluslararası Bilimsel Dergilerde Hakemlik
2. Journal of Statistical Computation and Simulation
3. IEEE Transactions on Signal Processing
4. Geo-Spatial Information Science
5. Plos One
6. IET Signal Processing
7. Concurrency and Computation: Practice and Experience
8. Wiley Interdisciplinary Reviews: Computational Statistics
9. Electronics Letters
10. Journal of Inequalities and Applications
11. Communications in Statistics - Simulation and Computation
12. Soft Computing
13. Iranian Journal of Science and Technology, Transactions A: Science
14. Journal of Public Affairs
15. Journal of Mathematics
16. Computational Intelligence and Neuroscience
17. Journal of Inverse and Ill-Posed Problems
18. Chemometrics and Intelligent Laboratory Systems
19. Heliyon
20. Computer Methods and Programs in Biomedicine
21. Big Data Research
22. İktisat Politikası Araştırmaları Dergisi
23. Adıyaman Üniversitesi Fen Bilimleri Dergisi
24. Bitlis Eren Üniversitesi Fen Bilimleri Dergisi
25. EKEV Akademi Dergisi
26. Erciyes Üniversitesi İktisadi ve İdari Bilimler Fakültesi Dergisi
27. Elektronik Sosyal Bilimler Dergisi
28. Fırat Üniversitesi Sosyal Bilimler Dergisi
29. Gazi University Journal of Science
30. Marmara Fen Bilimleri Dergisi
31. Mugla Journal of Science and Technology
32. Physical Sciences
33. Social Sciences
34. Bilecik Şeyh Edebali Üniversitesi Fen Bilimleri Dergisi
35. Mehmet Akif Ersoy Üniversitesi İktisadi ve İdari Bilimler Fakültesi Dergisi
36. Trakya Üniversitesi Sosyal Bilimler Dergisi
37. Politeknik Dergisi
38. International Journal of Advances in Engineering and Pure Sciences
39. Fırat Üniversitesi Sosyal Bilimler Dergisi